STATISTICS / INDIVIDUAL PROJECT

Individual Hypothesis-Testing Project

Is the average taxi trip in Armenia 14.4km?



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STUDENT BRIEF

The Hypothesis Test

Evaluating whether sample data provide sufficient evidence to challenge the 14.4 km average taxi trip distance in Armenia.

Testing the claim that the average taxi trip distance in Armenia is 14.4 km.

Null Hypothesis (H_0)

$H_0 = 14.4$ km

Alternative Hypothesis (H_1)

$H_1 \neq 14.4$ km

Our goal: Determine whether the sample provides sufficient statistical evidence that the actual average taxi trip distance differs from 14.4 km.

THE CORE IDEA

1 **Question**

Is the average taxi trip distance in Armenia significantly different from 14.4 km?

2 **Evidence**

Sample mean: 14.196 km

3 **Action**

Fail to reject the null hypothesis.

A clear sequence keeps the analysis focused

Move from the business context to the statistical decision, then explain what it means.

01

Define

Average Armenian taxi trip distance

02

Hypothesize

H_0: $\mu = 14.4$ km, H_1: $\mu \neq 14.4$ km

03

Collect

54 taxi rides, distance in km

04

Summarize

Mean 14.196 km, SD 9.755 km

05

Select

One-sample t-test, two-tailed (σ unknown)

06

Evaluate

$t = -0.152$, $p = 0.8796 > \alpha = 0.05$

07

Interpret

Fail to reject H_0; 14.4 km remains plausible

Show the evidence behind your decision

Use a significance level of α = 0.05 and report each required element clearly.

ANALYSIS OUTPUTS

Describe: Sample mean = 14.196 km, SD = 9.755 km, n = 54 rides

Check: Rides assumed independent; n = 53 is large enough for the t-test to hold

Report: Critical value ±2.0066, t = −0.1522, p-value = 0.8796

Explain: $|t| = 0.152 < 2.007$ and $p = 0.8796 > 0.05$, so we fail to reject H_0 with both approaches

PROJECT DETAILS

Population and variable:	Armenian taxi trips; trip distance (km)
Benchmark:	14.4 km
Null hypothesis:	μ = 14.4 km
Alternative hypothesis:	μ ≠ 14.4 km
Data source and sample size:	n = 54 rides
Selected test and justification:	One-sample t-test, two-tailed (σ unknown)
Critical value(s):	±2.0066 (df = 52, α = 0.05)
Test statistic:	t = −0.1522
p-value:	0.8796 > α = 0.05

A statistical decision is not the whole story

The sample gives no evidence that the average taxi trip differs from 14.4 km.

Type I error

Concluding the average trip differs from 14.4 km when it truly equals 14.4 km (risk = $\alpha = 0.05$).

Type II error

Missing a real difference: failing to reject H_0 when the true average is not 14.4 km.

Significance ≠ importance

Statistical significance does not automatically mean the effect is practically important.

Failing to reject H_0 does not prove that H_0 is true.

Acknowledge limitations and recommend a next step.

Statistical decision:	Fail to reject H_0 ($p = 0.8796 > 0.05$)	**Business implications:**	No evidence the mean differs from 14.4 km
Limitations:	n = 54; assumes a representative sample	**Recommended action:**	Treat 14.4 km as plausible; collect more data

Average Armenian taxi ride distance

A business question about whether the population mean ride distance matches a benchmark.

BENCHMARK

14.4 km

The claim concerns the average distance per ride across the population—not that every individual ride is 14.4 km.

NULL HYPOTHESIS

$H_0: \mu = 14.4$

ALTERNATIVE HYPOTHESIS

$H_1: \mu \neq 14.4$

Define the scope of your sample

Taxi service(s):	Yandex Go
Observation period:	01/2026-10/2026
Sampling method:	random
Number of rides:	54

Two files, one consistent result

The workbook holds the calculations; this presentation explains the decision to fail to reject H_0.

.XLSX

Excel workbook

- Source data:54 taxi trip distances (km)
- Descriptive statistics: mean 14.196 km, SD 9.755 km
- Test calculations: t = −0.1522, critical value ±2.0066
- Reproducible hypothesis-test calculations

.PPTX

PowerPoint presentation

- Hypotheses: H_0: μ = 14.4 km vs H_1: μ ≠ 14.4 km
- Data and method: 54 rides, one-sample t-test
- Key finding: fail to reject H_0
- Limitations and next step: small sample, collect more data

Submission location:

Google Sheets

File-naming convention:

.pptx

Presentation length:

7 minutes